description, adding a concrete implementation for its structure and algorithms (points **(c)** and **(d)**).

This is exactly the relationship between ***priority queues*** and ***heaps***. A priority queue is an abstract data structure, that can be implemented in many ways (including a sorted list). A heap is a concrete implementation of the priority queue using an array to hold elements, and specific algorithms to enforce invariants.

2.4.1 Priority Queue at work

Imagine you are provided with a priority queue. It can come from a third-party library, or from a standard library (many languages, like C++ or Scala, provide an implementation for priority queues in their standard container lib).

You don’t need to know the internals of the library at this point, you just need to follow its public API and use it, confident it’s properly implemented: this is the black box approach (figure 2.1).

For instance, let’s suppose we add our bugs to our PQ in the same order we have seen before:

Task Description	Severity (1-10)
Page loads take 2+ seconds	7
UI breaks on browser X	9
Optional form field blocked when using browser X on Friday the 13th	1
CSS style causes misalignment	8
CSS style causes 1px misalignment on browser X	5